

RAJNANDINI CHAUHAN

Software Developer & Full-Stack AI Systems — Frontend Engineering

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Education

Jaypee University of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering

Aug 2023 – Present

CGPA: 6.96 / 10

Projects

SkillSwap — AI-Powered Peer Learning Platform 🔗

Apr 2026 – Present

React.js, TypeScript, Tailwind CSS, Context API, Gemini API

- Developed a responsive peer-to-peer learning platform that enables users to exchange skills, discover learning partners, and manage personalized learning sessions.
- Integrated Gemini API to generate contextual quiz questions from user-provided notes and identify gaps in the learner's understanding.
- Architected the frontend using reusable React and TypeScript components with Context API for centralized state management.
- Designed responsive interfaces for skill discovery, peer profiles, session scheduling, quizzes, and learning progress across desktop and mobile devices.

License Plate Detection System 🔗

Jan 2025 – Present

Python, YOLOv8, OpenCV, FastAPI, React.js, NumPy

- Fine-tuned and optimized the model over 50 epochs, achieving **97.88% mAP@50**, **66.90% mAP@50–95**, **96.78% precision**, and **95.94% recall** on the validation set.
- Managed dataset preparation, image annotation, preprocessing, training configuration, and confidence-based model evaluation.
- Developed a FastAPI backend to process inference requests and return detection coordinates, confidence scores, and processed image results.
- Built a React dashboard for real-time visualization, search, filtering, and monitoring of license plate detection records.

Multimodal RAG for Research Paper Visual Question Answering

Ongoing

Python, Streamlit, ChromaDB, OpenAI API, PyMuPDF, Plotly, Pandas

- Developing a multimodal retrieval-augmented generation system to answer questions from research paper text, figures, tables, and captions with source citations.
- Building a PDF extraction pipeline using PyMuPDF and pdfplumber to process textual sections, images, captions, and structured table data.
- Creating a unified ChromaDB vector index for text chunks, table content, and generated image descriptions to support cross-modal semantic retrieval.
- Planning retrieval evaluation using Precision@K, Recall@K, MRR, answer correctness, and average response latency.

Technical Skills

Languages: Python, JavaScript, TypeScript, C++, SQL

Frontend: React.js, Tailwind CSS, CSS Modules, Context API, Responsive UI, Framer Motion, Streamlit

Backend: Node.js, Express.js, FastAPI, REST APIs

AI / ML: Multimodal RAG, Vector Embeddings, YOLOv8, OpenCV, Gemini API, OpenAI API, NumPy, Pandas

Databases / Retrieval: MongoDB, SQL, ChromaDB, Vector Search

Tools: Git, GitHub, VS Code, Linux, Vercel, Plotly

Problem Solving: Arrays, Strings, Two Pointers, Sliding Window, Sorting, Hashing, Basic DSA

Achievements

- SIH 2025 Hackathon Qualifier — Contributed to a face-recognition-based solution as part of a team.
- Cummins Scholar — Recognized for academic merit.
- Club Coordinator — Organized coding events and coordinated student team sessions.

Languages & Interests

Languages: English, Hindi

Interests: UI Design, Web Development, Hackathons, AI Research